

- 5 (a) Infectious diseases are caused by pathogens and are described as transmissible diseases.

Explain what is meant by the term transmissible.

.....
..... [1]

- (b) *Vibrio cholerae* is the bacterium that causes cholera.

Fig. 5.1 is a transmission electron micrograph of *V. cholerae*.

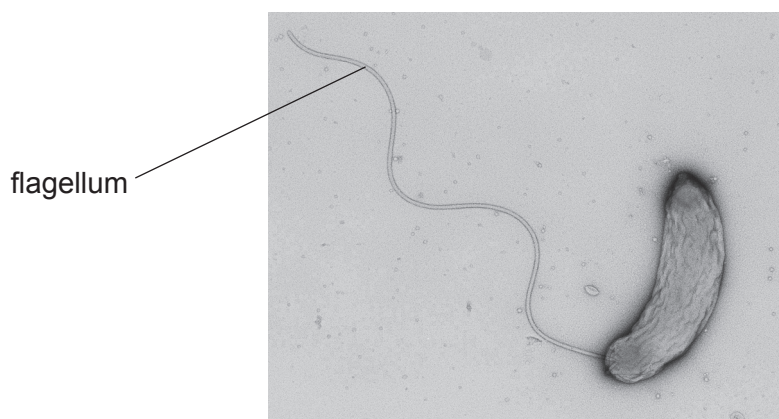


Fig. 5.1

A student wanted to know the actual length of the flagellum shown in Fig. 5.1.

State the information that is needed so that the student can calculate the actual length of the flagellum.

.....
.....
..... [2]

(c) Table 5.1 shows data on large outbreaks of cholera that occurred in seven countries in 2019.

The case-fatality rate is the number of deaths as a percentage of the number of cases.

Table 5.1

country	number of cases of cholera	number of deaths from cholera	case-fatality rate / %
Cameroon	1 307	55	4.2
DR Congo	29 087	501	1.7
Haiti	684	3	0.4
Kenya	5 150	39	0.8
Somalia	9 968	50	0.5
Sudan	346	11	3.2
Yemen	2 260 495	3767	

- (i) Complete Table 5.1 by calculating the case-fatality rate for Yemen. [1]
- (ii) Suggest why large outbreaks of cholera occurred in the countries listed in Table 5.1, but there were very few cases in most other countries.

[5]

[Total: 9]